使用Python

Astra Automation 21.12

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使用Python

NetApp Astra Control Python SDK

NetApp Astra Control Python SDK是開放原始碼套件、可用於自動化Astra Control部署。此套件也是瞭解Astra Control REST API的寶貴資源、可能是建立自己自動化平台的一部分。

> 為了簡化作業、NetApp Astra Control Python SDK將在本頁的其餘部分中稱為* SDK*。

兩種相關軟體工具

SDK包含兩種不同但相關的工具、可在存取Astra Control REST API時以不同的抽象層級運作。

Astra SDK

Astra SDK提供核心平台功能。其中包含一組Python類別、可抽象化基礎REST API呼叫。這些類別可支援各種Astra Control資源的管理動作、包括應用程式、備份、快照和叢集。

Astra SDK是套件的一部分、以單一的「astrasdk.py」檔案提供。您可以將此檔案匯入您的環境、並直接使用類別。

> * NetApp Astra Control Python SDK*（或僅SDK）是整個套件的名稱。* Astra SDK*是指單一檔案「astrasdk.py」中的核心Python類別。

工具套件指令碼

除了Astra SDK檔案之外、也提供「toolkit.py」指令碼。此指令碼可存取內部定義為Python功能的獨立管理動作、以更高的抽象層級運作。指令碼會匯入Astra SDK、並視需要呼叫類別。

如何存取

您可以使用下列方式存取SDK。

Python套件

SDK可從取得 "Python套件索引" 以* NetApp-Astra工具套件*的名稱。套件已指派版本編號、並將視需要繼續更新。您必須使用*子母畫面*套件管理公用程式、將套件安裝到您的環境中。

請參閱 "PyPI：NetApp Astra Control Python SDK" 以取得更多資訊。

GitHub原始程式碼

您也可以在GitHub取得SDK原始程式碼。儲存庫包含下列項目：

- 「astrasdk.py」（使用Python類別的Astra SDK）
- 「toolit.py」（較高層級的功能型指令碼）
- 詳細的安裝要求與指示

- 安裝指令碼
- 其他文件

您可以複製 "GitHub：NetApp/NetApp-Astra工具套件" 儲存庫。

安裝與基本需求

在安裝套件及準備使用套件時、需要考量幾種選項和需求。

安裝選項摘要

您可以使用下列其中一種方法來安裝SDK：

- 使用Pip將套件從PyPI安裝到Python環境
- 複製GIT Hub儲存庫、然後：
  - 將套件部署為Docker容器（包含您所需的一切）
  - 複製兩個核心Python檔案、以便Python用戶端程式碼存取

如需詳細資訊、請參閱PyPI和GitHub頁面。

Astra Control環境的需求

無論是直接使用Astra SDK中的Python類別、或是「toolkit.py」指令碼中的功能、最終都能在Astra Control部署中存取REST API。因此、您需要Astra帳戶和API權杖。請參閱 "開始之前" 如需詳細資訊、請參閱本文件*入門*一節中的其他頁面。

NetApp Astra Control Python SDK的需求

SDK有幾項與本機Python環境相關的先決條件。例如、您必須使用Python 3.5或更新版本。此外、還需要幾個Python套件。如需詳細資訊、請參閱GitHub儲存庫頁面或PyPI套件頁面。

實用資源摘要

以下是您開始使用所需的資源摘要。

- "PyPI：NetApp Astra Control Python SDK"
- "GitHub：NetApp/NetApp-Astra工具套件"

原生Python

開始之前

Python是熱門的開發語言、尤其是資料中心自動化。在將Python的原生功能與數個通用套件搭配使用之前、您必須先準備環境和所需的輸入檔。

> 除了直接使用Python存取Astra Control REST API之外、NetApp也提供工具套件、可擷取API並移除部分複雜性。請參閱 "NetApp Astra Control Python SDK" 以取得更多資訊。

準備環境

執行Python指令碼的基本組態需求如下所述。

Python 3.

您需要安裝最新版本的Python 3。

其他程式庫

必須安裝*請求*和* urllib3*程式庫。您可以根據環境使用pip或其他Python管理工具。

網路存取

執行指令碼的工作站必須具備網路存取權、而且必須能夠連線到Astra Control。使用Astra Control Service時、您必須連線至網際網路、才能連線至https://astra.netapp.io`上的服務。

身分識別資訊

您需要具有帳戶識別碼和API權杖的有效Astra帳戶。請參閱 "取得API權杖" 以取得更多資訊。

建立**Json**輸入檔

Python指令碼仰賴Json輸入檔中包含的組態資訊。範例檔案如下所示。

您需要視環境的需求更新範例。

身分識別資訊

下列檔案包含API權杖和Astra帳戶。您需要使用「-I」（或「-identity」）CLI參數、將此檔案傳遞給Python指令碼。

```
{
  "api_token": "kH4CA_uVIa8q9UuPzhJaAHaGlaR7-no901DkkrVjIXk=",
  "account_id": "5131dfdf-03a4-5218-ad4b-fe84442b9786"
}
```

列出託管應用程式

您可以使用下列指令碼來列出Astra帳戶的託管應用程式。

請參閱 "開始之前" 以取得所需Json輸入檔的範例。

```
#!/usr/bin/env python3
##-----------------------------------------------------------------------
------
#
```

```
# Usage: python3 list_man_apps.py -i identity_file.json
#
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#
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#
##-----------------------------------------------------------------------
------

import argparse
import json
import requests
import urllib3
import sys

# Global variables
api_token = ""
account_id = ""

def get_managed_apps():
    ''' Get and print the list of managed apps '''

    # Global variables
    global api_token
    global account_id

    # Create an HTTP session
    sess1 = requests.Session()

    # Suppress SSL unsigned certificate warning
    urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

    # Create URL
    url1 = "https://astra.netapp.io/accounts/" + account_id +
"/k8s/v1/managedApps"

    # Headers and response output
    req_headers  = {}
```

```
    resp_headers = {}
    resp_data    = {}

    # Prepare the request headers
    req_headers.clear
    req_headers['Authorization'] = "Bearer " + api_token
    req_headers['Content-Type'] = "application/astra-managedApp+json"
    req_headers['Accept'] = "application/astra-managedApp+json"

    # Make the REST call
    try:
        resp1 = sess1.request('get', url1, headers=req_headers,
allow_redirects=True, verify=False)

    except requests.exceptions.ConnectionError:
        print("Connection failed")
        sys.exit(1)

    # Retrieve the output
    http_code = resp1.status_code
    resp_headers = resp1.headers

    # Print the list of managed apps
    if resp1.ok:
        resp_data = json.loads(resp1.text)
        items = resp_data['items']
        for i in items:
            print(" ")
            print("Name: " + i['name'])
            print("ID: " + i['id'])
            print("State: " + i['state'])
    else:
        print("Failed with HTTP status code: " + str(http_code))

    print(" ")

    # Close the session
    sess1.close()

    return

def read_id_file(idf):
    ''' Read the identity file and save values '''

    # Global variables
    global api_token
    global account_id
```

```
    with open(idf) as f:
        data = json.load(f)

    api_token = data['api_token']
    account_id = data['account_id']

    return

def main(args):
    ''' Main top level function '''

    # Global variables
    global api_token
    global account_id

    # Retrieve name of JSON input file
    identity_file = args.id_file

    # Get token and account
    read_id_file(identity_file)

    # Issue REST call
    get_managed_apps()

    return

def parseArgs():
        ''' Parse the CLI input parameters '''

        parser = argparse.ArgumentParser(description='Astra REST API -
List the managed apps',
                            add_help = True)
        parser.add_argument("-i", "--identity", action="store", dest
="id_file", default=None,
                            help='(Req) Name of the identity input file',
required=True)

        return parser.parse_args()

if __name__ == '__main__':
    ''' Begin here '''

    # Parse input parameters
    args = parseArgs()

    # Call main function
```

```
    main(args)
```

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